A. LongSciVerify: Abstractive Summarisation Methods

For our human evaluation, we provide summaries generated using three different abstractive summarisation methods, which all consider text from across the entire length of a long document when generating a summary.

Text zoning is a task which aims to segment a larger body of text into different zones or sections (Teufel et al., 1999). Two of the methods we implement apply text zoning, and treat the identified sections independently, so that a PLM used to train the abstractive summarisation model is only required to process a document section at a time, rather than the entire document, to avoid truncation. We first implement DANCER (Gidiotis and Tsoumakas, 2020), a SOTA method which fine-tunes the PEGASUS PLM (Zhang et al., 2020a). DANCER splits a document into zones using keyword matching, then finds corresponding sections of the target abstract using ROUGE matching (Lin, 2004). It then uses beam search decoding to generate the summaries and combines the generated summaries of each section to form the article summary. An example output of the DANCER method can be seen in the top block of Figure 5. The second method we implement is a method we develop. It applies a similar zoning approach to DANCER, but with two notable differences. Firstly, the sections of the summary used to create the target training pairs are matched by using both keywords and assumptions about the structure of the long scientific documents used in our study (Cohan et al., 2018). Consequently, the target sentences for each section are always sentences which were consecutive to one another in the original summary, thus resulting in a summary which follows the logical structure of the document. Secondly, the summaries generated do not use beam decoding and are highly structured as each section of the summary is prefixed with the type of section it was generated from, e.g., '*results:*'. An example of a summary generated by this method can be seen in the middle block of Figure 5. The last method we implement uses an extractive-abstractive approach. We train our extractive-abstractive model using ORACLE extractive summaries as an input, optimized for a recall metric but limiting the number of sentences selected so that the total number of input tokens is less than 1024. At test time, we implement the unsupervised, extractive method GenCompareSum (Bishop et al., 2022), which does not truncate the source document and has previously shown strong performance on the PubMed and ArXiv data sets. An example of a summary generated with this method can be seen in the final block of Figure 5. We use the train/val/test splits from the original data sets and use DANCER and GenCompareSum with their default settings. For our zoning method and the extractive-abstractive method, we fine-tune the LED PLM (Beltagy et al., 2020) for three epochs with its default parameters. All experiments are run on a single NVIDIA v100 GPU.

background : urinary tract infections (utis) are one of the inflammatory diseases produced by high multiplication of many pathogens in the urinary apparatus , resulting in alterations in the perfect function of the urinary tract and kidneys . uropathogenic escherichia coli (upec) strains have special virulence factors , including pili or fimbriae , which mediate attachment to uroepithelial and vaginal cells , resistance to human serum bactericidal activity , haemolysin production , and increased amounts of k capsular antigen . furthermore , virulence factors of upec strains the fimh gene was found in 130 isolates (92.8%) of upec . of the 130 isolates positive for the fimh gene , 62 (47.7%) and 68 (52.3%) belonged to hospitalized patients and outpatients , respectively . the aim of this study was to determine the prevalence of the fimh gene in urothelial epithelial cells (upecs) isolated from hospitalized patients and outpatients with urinary tract infections (utis) referred to educational hospitals of shahrekord , iran.materials and methods : in this cross - sectional study ,130 upec isolates were isolated from hospitalized patients and outpatients with utis referred to educational hospitals of shahrekord , iran .

introduction: background urinary tract infections (utis) are one of the major causes of morbidity and mortality in many parts of the world. e. coli is one of the most frequent pathogen responsible for up to 80% of utis. objectives the aim of this study was to determine the virulence factors of type 1 fimbriae (fimh) isolated from uropathogenic e. coli isolates.

methods: : one hundred and six e. coli strains were collected from hospitalized and non - hospitalized patients with uti during 2012. the isolates were characterized and identified using gram staining and biochemical tests . genomic dna templates for pcr amplification were gained from overnight growth of bacterial isolates on luria - bertani agar and subjected to screening for the presence of the fimh gene by pcr .

results: : a total of 130 upec isolates were detected in the study . the fimh gene was detected in 92.8 % of upec isolates . the fimh gene was detected in 47.7 % of hospitalized patients and 52.3 % of outpatients . the high binding capacity of fimh was found to be in more than 95 % of the isolates .

background : in recent years, there has been a much increase in the incidence of urinary tract infection (upec) in iran. we evaluated the virulence of the fimh gene in upec isolates from hospitalized patients and out - patients with upec. methods : we investigated the fimh gene in the upec isolates from hospitalized patients and out - patients with uti referred to the educational hospitals of shahrekord, iran. the bacterial isolates were subjected to screening for the presence of the fimh gene by pcr. in addition to analyzing all the upec isolates, the fimh gene was detected in other strains of e. coli. results : of the 130 upec isolates studied, the fimh gene was found in 92.8% of upec isolates. the fimh gene was detected in 98% of the e. coli isolates from uti. the fimh gene was detected in 98% of upec isolates from a patient with uti. for more subsequent investigations, the fimh gene was detected in upec strains isolated from hospitalized and out - patients with uti

Figure 5: Automatically generated summaries included in our human evaluation study for one article sampled from the PubMed data set.